

《移动终端软件开发技术》

Mobile Application Development

8. Android网络通信

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1 Android WebView

1 WebView的用法

```
class MainActivity : ComponentActivity() {  
    override fun onCreate(savedInstanceState: Bundle?) {  
        super.onCreate(savedInstanceState)  
        enableEdgeToEdge()  
        // 设置窗口内容不延伸到系统窗口区域  
        WindowCompat.setDecorFitsSystemWindows(window, true)  
        setContent {  
            CWebViewTestTheme {  
                Surface(modifier = Modifier.fillMaxSize()) {  
                    Greeting(url = "https://www.bupt.edu.cn")  
                }  
            }  
        }  
    }  
}
```

```
@SuppressLint("SetJavaScriptEnabled")  
@Composable  
fun Greeting(url: String) {  
    Box(modifier = Modifier.fillMaxSize()) {  
        AndroidView(  
            modifier = Modifier.fillMaxSize(),  
            factory = { context ->  
                // 创建一个WebView实例  
                WebView(context).apply {  
                    // 设置WebViewClient, 用于处理页面加载事件  
                    webViewClient = WebViewClient()  
                    // 启用JavaScript支持  
                    settings.javaScriptEnabled = true  
                    // 加载指定的URL  
                    loadUrl(url)  
                }  
            }  
        )  
    }  
}
```

1 WebView的用法

```
<uses-permission android:name="android.permission.INTERNET" />
```

```
<application
    android:allowBackup="true"
    android:dataExtractionRules="@xml/data_extraction_rules"
    android:fullBackupContent="@xml/backup_rules"
    android:icon="@mipmap/ic_launcher"
    android:label="@string/app_name"
    android:roundIcon="@mipmap/ic_launcher_round"
    android:supportsRtl="true"
    android:theme="@style/Theme.CWebViewTest"
    android:usesCleartextTraffic="true"
    tools:targetApi="31">
    <activity
        android:name=".MainActivity"
        android:exported="true"
        android:label="@string/app_name"
        android:theme="@style/Theme.CWebViewTest">
        <intent-filter>
            <action android:name="android.intent.action.MAIN" />

            <category android:name="android.intent.category.LAUNCHER" />
        </intent-filter>
    </activity>
</application>
```



北邮要闻



2025.01.24

《人民日报》整版刊发2024年校长论坛发言摘编



2025.01.23

校领导春节前开展走访慰问



2025.01.22

学校举办“情满邮苑”留校学生新春慰问活动

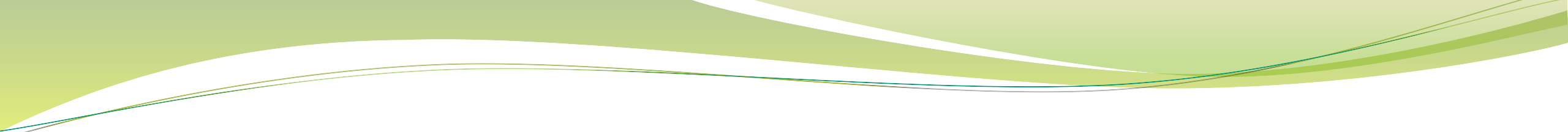


2025.01.21

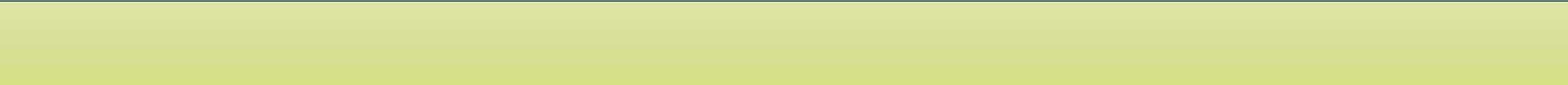
北京邮电大学大学生艺术团在2024年北京大学生音乐节中斩获佳绩

热点专题





2 Android HTTP



2 OkHttp

```
[versions]
# 定义 OkHttp 版本
okhttp = "4.11.0"

[libraries]
# 定义 OkHttp 库的依赖
okhttp = { group = "com.squareup.okhttp3", name = "okhttp", version.ref = "okhttp" }
```

```
dependencies {
    // 引用 OkHttp 库
    implementation(libs.okhttp)
}
```

1. `val client = OkHttpClient()`

1. `val request = Request.Builder().build()`

1. `val request = Request.Builder()`
2. `.url("https://www.baidu.com")`
3. `.build()`

1. `val response = client.newCall(request).execute()`

1. `val responseData = response.body?.string()`

2 OkHttp

```
1. val requestBody = FormBody.Builder()  
2.     .add("username", "admin")  
3.     .add("password", "123456")  
4.     .build()
```

```
1. val request = Request.Builder()  
2.     .url("https://www.baidu.com")  
3.     .post(requestBody)  
4.     .build()
```

```
1. <uses-permission android:name="android.permission.INTERNET" />
```

2 OkHttp

```
class MainActivity : ComponentActivity() {  
    override fun onCreate(savedInstanceState: Bundle?) {  
        super.onCreate(savedInstanceState)  
        enableEdgeToEdge()  
        WindowCompat.setDecorFitsSystemWindows(window, true)  
        setContent {  
            CNetworkTestTheme {  
                Surface(  
                    modifier = Modifier.fillMaxSize(),  
                    color = MaterialTheme.colorScheme.background  
                ) {  
                    Greeting()  
                }  
            }  
        }  
    }  
}
```

```
@Composable  
fun Greeting(modifier: Modifier = Modifier) {  
    // 创建一个垂直滚动状态  
    val scrollState = ScrollState(0)  
    // 子线程信息处理  
    var response by remember {mutableStateOf("The Default Response")}  
    val handler = Handler(Looper.getMainLooper())  
  
    // 使用 Column 结合 verticalScroll 修饰符实现滚动  
    Column(  
        modifier = modifier  
            .fillMaxWidth()  
            .verticalScroll(scrollState),  
        horizontalAlignment = Alignment.CenterHorizontally,  
        verticalArrangement = Arrangement.spacedBy(8.dp)  
    ) {  
        Button(  
            onClick = {  
                thread {  
                    handler.post {  
                        response = ""  
                        这是一段很长的文本，用于测试滚动功能。  
                        它可以包含多行内容，并且会根据文本框高度自动滚动显示。  
                        你可以不断添加更多的文本，直到文本超出显示范围。  
                        这样就可以看到滚动条的效果了。  
                        """".trimIndent()  
                    }  
                }  
            }  
        ) {  
            Text("Get the Response")  
        }  
        // 显示文本  
        Text(  
            text = response,  
            style = MaterialTheme.typography.bodyLarge,  
            textAlign = TextAlign.Start  
        )  
    }  
}
```


2 OkHttp

```
package com.example.cnetworktest

import okhttp3.*
import java.net.URL

object OkHttpTest {
    private val client = OkHttpClient()

    fun run(address: String): String? {
        try {
            val url = URL(address)
            val request = Request.Builder()
                .url(url)
                .build()
            val response = client.newCall(request).execute()
            val responseData = response.body?.string()
            return responseData
        } catch (e: Exception) {
            return e.toString()
        }
    }
}
```

```
@Composable
fun Greeting(modifier: Modifier = Modifier) {
    .....
    Column(.....) {
        Button(
            onClick = {
                thread {
                    // 在子线程中执行网络请求
                    val responseData = OkHttpTest
                        .run("https://www.bupt.edu.cn")
                        .toString()
                    // 使用 handler 在主线程更新 UI
                    handler.post {
                        response = responseData
                    }
                }
            }
        ) {.....}
        Text(.....)
    }
}
```

2 OkHttp

中国联通 | Club 0.00K/s

Get the Response

The Default Response

12:10

中国联通 | Club 158B/s

12:15

```
<!DOCTYPE html>
```

```
<html lang="en-US"><head>
```

```
<meta charset="UTF-8">
```

```
<meta name="description" content="">
```

```
<meta http-equiv="X-UA-Compatible" content="IE=edge">
```

```
<meta name="viewport" content="width=device-width,
initial-scale=1, shrink-to-fit=no">
```

```
<title>北京邮电大学</title><META Name="keywords"
Content="北京邮电大学" />
```

```
<link href="css/style.css" rel="stylesheet" type="text/
css">
```

```
<link href="css/mystyle.css" rel="stylesheet">
```

```
<link href="css/swiper.min.css" rel="stylesheet">
```

```
<link href="css/slick.css" rel="stylesheet">
```

```
<script type="text/javascript" src="js/bdtxkmin.js"></
script>
```

```
<script type="text/javascript">
```

```
///*子菜单*/
```

```
var SPAN = 103;
```

```
var width = $(window).width();
```

```
var height = $(window).height();
```

```
$(function() {
```

```
    if (width >= 1680) {
```

```
        SPAN = 124;
```

```
    } else {
```

```
        SPAN = 103;
```

```
    }
```

```
    if (height < 550) height = 550;
```

3 网络请求回调

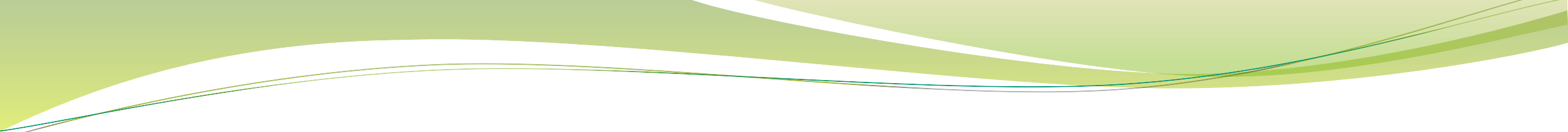
3 网络请求回调

```
interface Callback {  
    fun onResponse(call: Call, response: Response)  
    fun onFailure(call: Call, e: IOException)  
}
```

```
import okhttp3.Callback  
import okhttp3.OkHttpClient  
import okhttp3.Request  
  
object OkHttpTest {  
  
    private val client = OkHttpClient()  
  
    fun run(address: String, callback: Callback) {  
        val request = Request.Builder()  
            .url(address)  
            .build()  
  
        client.newCall(request).enqueue(callback)  
    }  
}
```

3 网络请求回调

```
.....  
// 子线程信息处理  
var responseData by remember {mutableStateOf("The Default Response")}  
val handler = Handler(Looper.getMainLooper())  
  
Column(.....) {  
    Button(  
        onClick = {  
            OkHttpTest.run("https://www.bupt.edu.cn",  
                object : Callback {  
                    override fun onResponse(call: Call, response: Response) {  
                        try {  
                            if (response.isSuccessful) {  
                                val responseString = response.body?.string()  
                                handler.post {  
                                    responseData = responseString ?: "Response body is null"  
                                }  
                            } else {  
                                handler.post {  
                                    responseData = "HTTP错误: CODE ${response.code}"  
                                }  
                            }  
                        } catch (e: Exception) {  
                            handler.post {  
                                responseData = "其他错误: ${e.message}"  
                            }  
                        }  
                    }  
                    override fun onFailure(call: Call, e: IOException) {  
                        handler.post {  
                            responseData = "网络错误: ${e.message}"  
                        }  
                    }  
                })  
        })  
    }  
)  
Text(.....)  
}
```



感谢观看！

